

SMD LED



Lead-Free Parts

LG-1309-9UG-CT-A

DATA SHEET

DOC.NO : IMQW0905-LG-1309-9UG-CT-A

REV. : A

DATE : 31 – Oct. - 2025

Features:

1. Package in 12mm carrier tape on 7" diameter reel.
2. Compatible with automatic placement equipment.
3. Compatible with infrared and vapor phase reflow solder process.
4. Multicolor type.
5. Pb-free.
6. The product itself will remain within RoHS compliant version.

Descriptions:

1. The LG-1309 SMD Taping is much smaller than lead frame type components, thus enable smaller board size, higher packing density, reduced storage space and finally smaller equipment to be obtained.
2. Besides, lightweight makes them ideal for miniature applications. etc.

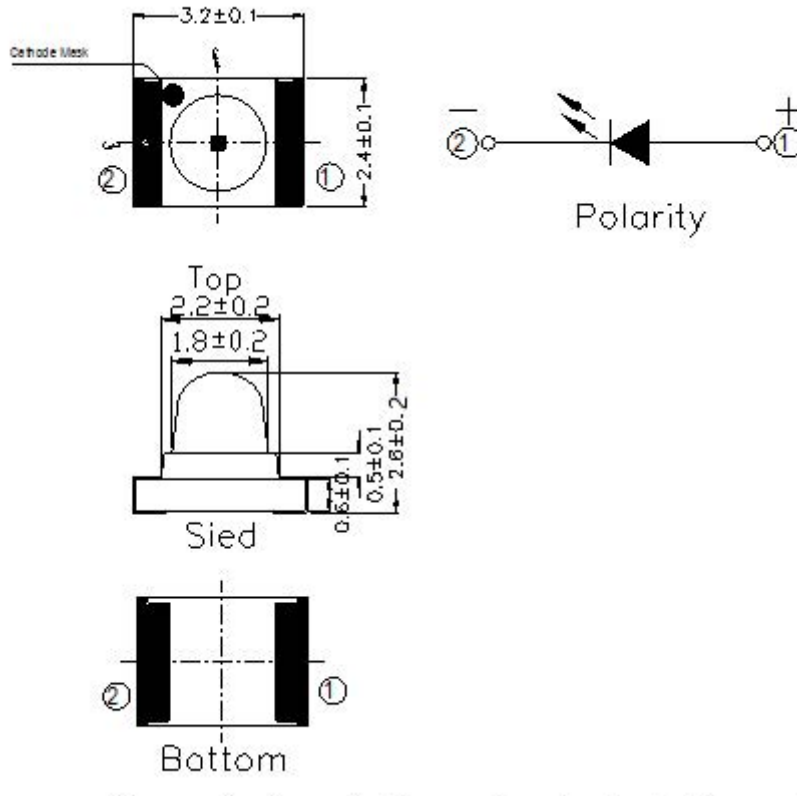
Applications:

1. Backlighting in dashboard and switch.
2. Telecommunication: indicator and backlighting in telephone and fax.
3. Flat backlight for LCD, switch and symbol
4. General use.

Device Selection Guide

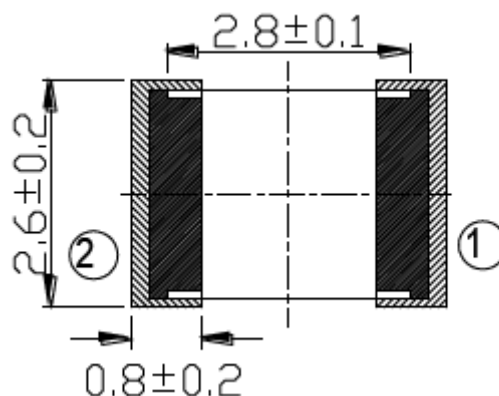
Part No.	Material	COLOR	
		Emitted	Lens
LG-1309-9UG-CT-A	AlGaInP	Yellow Green	Water Clear

Package Outline Dimensions



Note : 1.All dimension are in millimeter tolerance is ± 0.1 mm unless otherwise noted.
2.Specifications are subject to change without notice.

Recommended Soldering Pad Dimensions



Note : The tolerances unless mentioned is ± 0.1 mm
Suggested pad dimension is just fo reference only.
Please modify the pad dimension based on individual need.

Absolute Maximum Ratings at Ta=25°C

Parameter	Symbol	Ratings	UNIT
Power Dissipation	PD	60	mW
Peak forward current(Duty 1/10@1KHz)	I _{FP}	60	mA
Forward Current Per Chip	I _F	25	mA
Reverse Voltage	V _r	5	V
Electrostatic Discharge	ESD (HBM)	2000	V
Operating Temperature	T _{opr}	-40 ~ +85	°C
Storage Temperature	T _{stg}	-40 ~ +90	°C
Soldering Temperature	T _{sol}	Reflow Soldering : 260 °C for 10 sec. Hand Soldering : 350 °C for 3 sec.	

Typical Electrical & Optical Characteristics (Ta=25°C)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
Luminous Intensity	IV	715	----	1800	mcd	IF=20mA
Peak Wavelength	λ _p	----	575	----	nm	
Dominant Wavelength	λ _d	568.5	----	573.0	nm	
Spectrum Radiation	△λ	----	20	----	nm	
Forward Voltage	V _F	1.7	----	2.3	V	
Viewing Angle	2θ1/2	----	20	----	Deg	V _r =5V
Reverse Current	I _r	----	----	10	μA	

Note : 1.The forward voltage data did not including ±0.05V testing tolerance.

2.The luminous intensity data did not including ±11% testing tolerance.

3.The Dominant Wavelength data did not including ±1nm

4..Reverse Voltage(V_R) Condition is applied to IR test only The device is not designed for reverse operation

Luminous Intensity Classification

BIN CODE	IV (mcd) at 20mA	
	Min	Max
V1	715	900
V2	900	1120
W1	1120	1420
W2	1420	1800

Dominant Wavelength Classification

BIN CODE	λd (nm) at 20mA	
	Min	Max
CC1	568.5	570
CC2	570	571.5
CC3	571.5	573

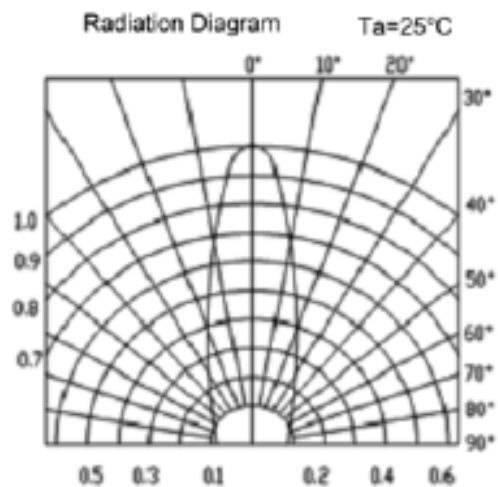
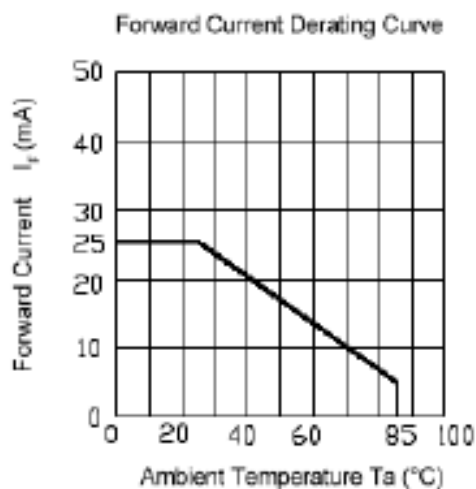
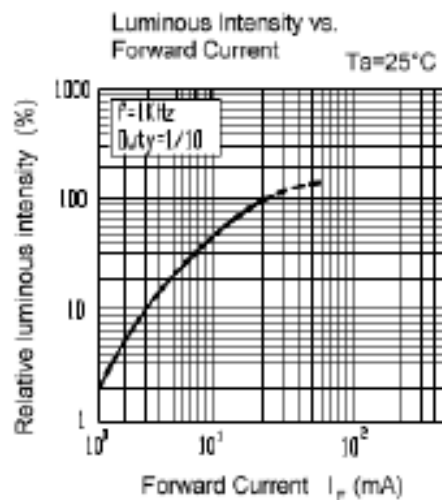
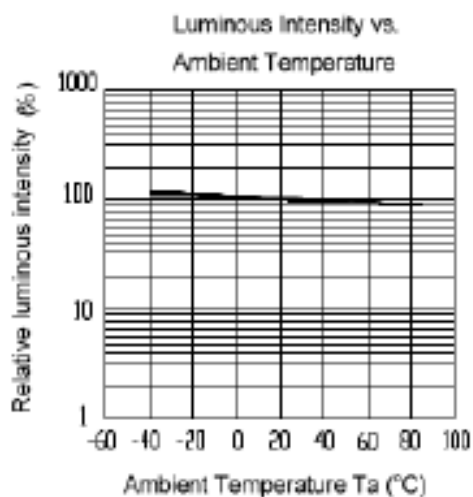
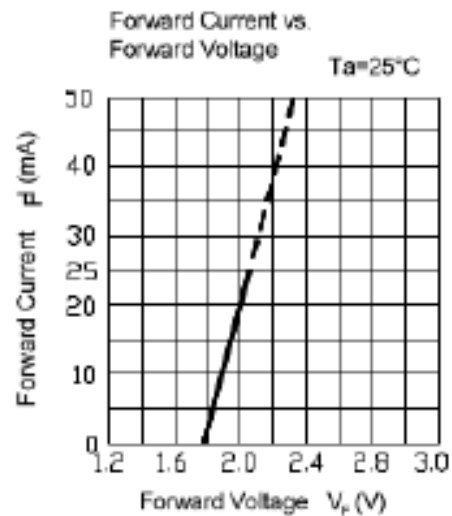
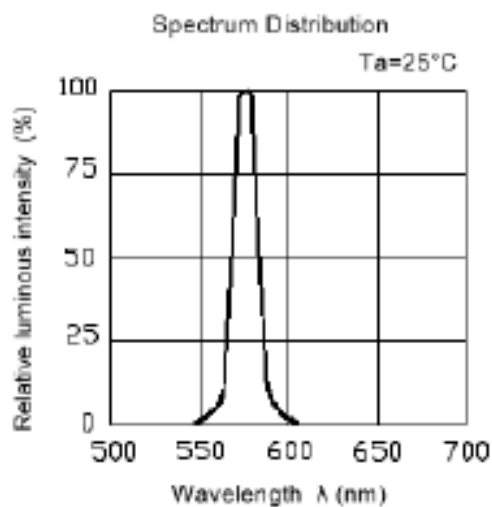
Forward Voltage Classification

BIN CODE	VF(V) at 20mA	
	Min	Max
19	1.7	1.8
20	1.8	1.9
21	1.9	2.0
22	2.0	2.1
23	2.1	2.2
24	2.2	2.3

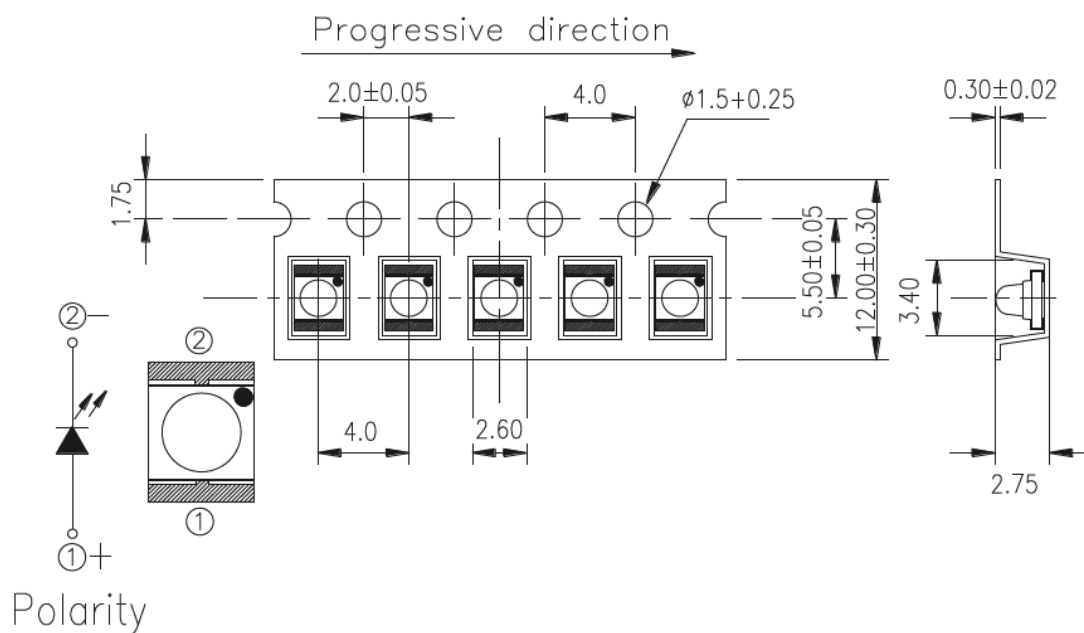
Note:

1. Tolerance of Luminous Intensity $\pm 11\%$
2. Tolerance of The Dominant Wavelength $\pm 1\text{nm}$
3. Tolerance of The Forward Voltage $\pm 0.05\text{V}$

Typical Electro-Optical Characteristics Curve

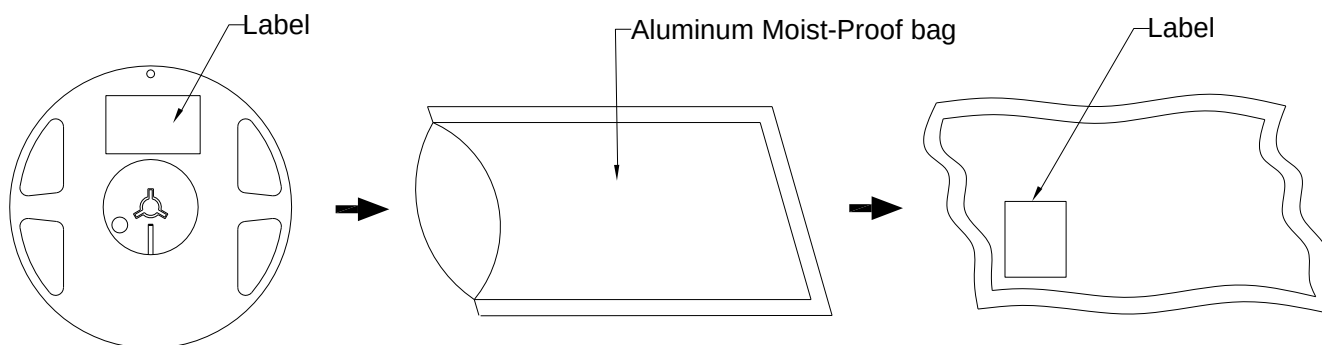


Carrier Tape Dimensions



Note : The tolerances unless mentioned is ± 0.1 mm, Angle ± 0.5 . Unit=mm.

Packing Specifications



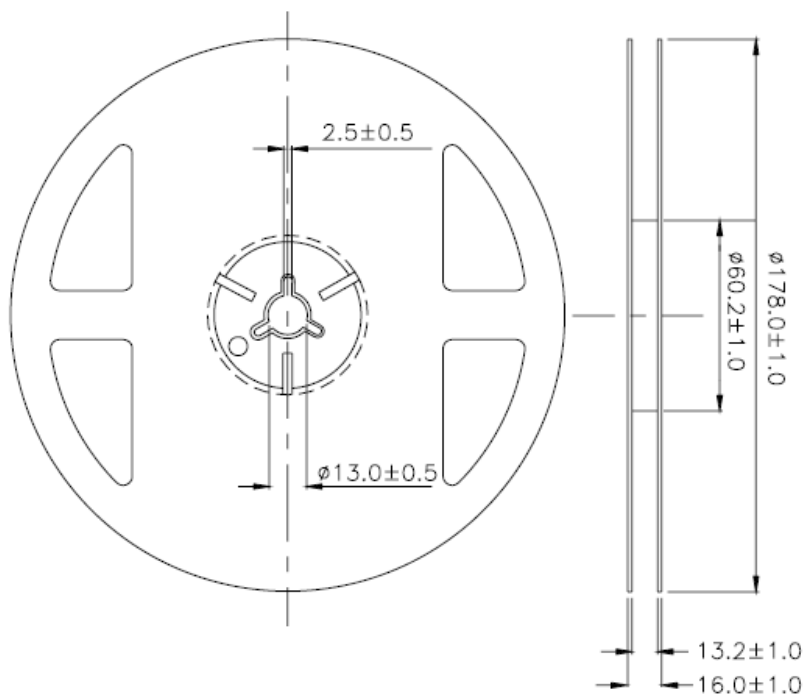
Part No.	Description	Quantity/Reel
LG-1309-9UG-CT-A	8.0mm tape, 7" reel	1000 devices

Label Explanation

		LIGITEK ELECTRONICS CO., LTD.	
			
PART:	LG-1309-9UG-CT-A		
			
LOT:	GS122090168		
			
QTY(PCS):	XXXX		
			
BIN/HUE:	XX		
		VF:XX-XX	

BIN: : Luminous Intensity
HUE: Dominant Wavelength
VF: Forward Voltage

Reel Dimensions



Precautions For Use:

Storage time:

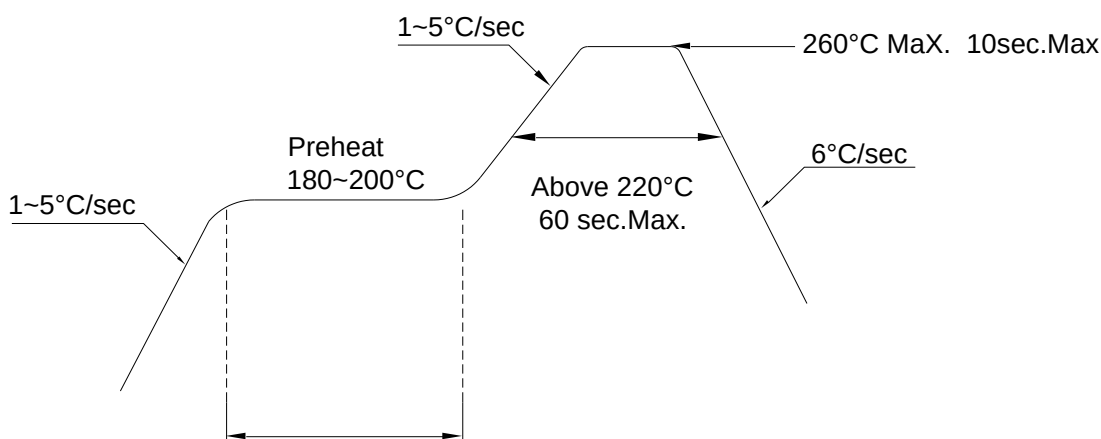
1. Calculated shelf life before opening is 18 months at $< 30^{\circ}\text{C}$ and $< 90\%$ relative humidity (RH)
2. After bag is opened, devices which will be subjected to reflow soldering or other high temperature processes must be
 - a) Assembled within one years in an environment of $\leq 30^{\circ}\text{C}$ / 60% RH, or
 - b) Stored at ambient of 10% RH or less
3. Devices are required baking before assembly if:
 - a) Humidity Indicator Card reads $>10\%$ (for level 2a -5a) or $>60\%$ (for level 2) at ambient temperature $23\pm 5^{\circ}\text{C}$
 - b) 2.a) or 2.b) doesn't meet
4. If baking is required, devices should be baked for >72 hours at $60\pm 5^{\circ}\text{C}$ / 5% RH. Performing baking only once, and using the baked devices within 72 hours.

MSL 2

Over current- protection

The LEDs is sensitive parts, slight voltage shift will cause big change and will cause burn out. Customer must apply resistors for protection.

LED soldering



Note:

1. Reflow soldering should not be done more than two times.
2. When soldering, do not put stress on the LEDs during heating.
3. After soldering, do not warp the circuit board.

Repairing

In principle repair should not be done after the LEDs have been soldered. When repairing is unavoidable, it should be confirmed before hand not to be damaged whether the characteristics of the LEDs by repairing and a double-head soldering iron should be used (as below description figure).

